

Supplement to “The Complexity of Stationary Diffusion-Driven Instability in Mass-Action Reaction Networks”

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1 Purpose and independence protocol

This supplement records the Phase III proof reconstruction and validation artifacts. The proofs below were rewritten from definitions and are organized in dependency order. The independent implementation under `independent_verifier/` imports no code from the inherited candidate package. Agreement with the inherited implementation was treated as a regression check only; exact proofs and definition-level tests are the basis for the validation outcome.

The central scope is weak stationary linear instability: some nonzero spatial mode has a positive real eigenvalue. No first-crossing, wave-exclusion, or nonlinear pattern claim is introduced in this supplement.

2 Diagonal damping at a fixed Jacobian

For a real matrix $J \in \mathbb{R}^{n \times n}$ and a set $I \subseteq [n]$, define

$$c_I(J) = (-1)^{|I|} \det J[I, I], \quad c_\emptyset(J) = 1.$$

No stability assumption is made in the next theorem.

Theorem 2.1 (Positive diagonal damping). *The following statements are equivalent.*

1. *There is a diagonal matrix $D \succ 0$ for which $J - D$ has a strictly positive real eigenvalue.*
2. *$c_I(J) < 0$ for at least one $I \subseteq [n]$.*

When J is Hurwitz, every set in item 2 is nonempty and proper.

Proof. Write $D(d) = \text{diag}(d_1, \dots, d_n)$. Multilinearity of the determinant in its columns gives

$$\det(D(d) - J) = \sum_{I \subseteq [n]} (-1)^{|I|} \det J[I, I] \prod_{j \notin I} d_j. \quad (1)$$

Indeed, choosing the diagonal column from $D(d)$ at every index in I^c leaves the principal block $-J[I, I]$ after the corresponding rows and columns are deleted. The surviving sign is $(-1)^{|I|}$. This also covers $I = \emptyset$, whose contribution is $\prod_j d_j$, and $I = [n]$, whose contribution is $\det(-J)$.

Suppose first that every $c_I(J)$ is nonnegative. For arbitrary $D \succ 0$ and $\lambda > 0$, apply (1) to $d_i + \lambda$. Every summand is nonnegative and the empty-set summand is strictly positive. Therefore

$$\det(\lambda I - (J - D)) = \det(D + \lambda I - J) > 0.$$

A positive real eigenvalue would make this determinant zero, so item 1 is impossible. This proves the necessity of item 2 by contraposition.

Now assume $c_I(J) < 0$. For $t > 1$ set

$$d_j(t) = \begin{cases} t, & j \notin I, \\ t^{-1}, & j \in I. \end{cases}$$

The power of t in the monomial indexed by K in (1) is

$$e_I(K) = |(I^c \cap K^c)| - |I \cap K^c|.$$

For the chosen set I , $e_I(I) = n - |I|$, and direct cancellation gives

$$e_I(I) - e_I(K) = |I \triangle K|.$$

Thus the I -term is the unique term of highest weight. Dividing by $t^{n-|I|}$ and sending t to infinity yields

$$t^{-(n-|I|)} \det(D(d(t)) - J) \longrightarrow c_I(J) < 0.$$

Hence some finite t gives $\det(D(d(t)) - J) < 0$, with every diagonal entry of $D(d(t))$ strictly positive. The characteristic polynomial

$$p(\lambda) = \det(\lambda I - (J - D(d(t))))$$

is monic, is negative at $\lambda = 0$, and tends to $+\infty$ as $\lambda \rightarrow +\infty$. The intermediate value theorem therefore supplies a positive real zero.

If J is Hurwitz, then $c_\emptyset = 1$. Also $c_{[n]} = \det(-J) > 0$, because it is the constant coefficient of the monic Hurwitz characteristic polynomial. Neither endpoint set can witness item 2. The argument did not assume full rank; a zero full determinant or any other degenerate principal minor merely contributes a zero coefficient. $\square$

Remark 2.2 (Constructive witness). The proof is effective over $\mathbb{Q}$. Once a negative signed principal minor is specified, test $t = 2, 4, 8, \dots$ until the exact determinant becomes negative. Termination follows from the unique-leading-term limit.

2.1 Designated mobile coordinates

Let $M \subseteq [n]$ and let $D_M(d)$ have positive diagonal entries on M and zeros on M^c .

Theorem 2.3 (Positive-spectral-parameter mobile criterion). *Assume*

$$\det(\lambda I - J) > 0 \quad \text{for every } \lambda > 0.$$

Then a matrix $D_M(d)$ exists for which $J - D_M(d)$ has a positive real eigenvalue if and only if there are $\lambda > 0$ and $I \supseteq M^c$ such that

$$\det(\lambda I_{|I|} - J[I, I]) < 0.$$

Proof. For fixed $\lambda > 0$, expansion only in the mobile diagonal variables gives

$$\det(\lambda I - J + D_M(d)) = \sum_{I \supseteq M^c} \det(\lambda I_{|I|} - J[I, I]) \prod_{j \in M \setminus I} d_j. \quad (2)$$

If the left side vanishes, the coefficient indexed by $[n]$ is positive by (2.3); therefore at least one other coefficient must be negative.

Conversely, fix a negative coefficient indexed by I . Assign $d_j = t$ for $j \in M \setminus I$ and $d_j = t^{-1}$ for $j \in M \cap I$. The exponent gap between the I -monomial and the K -monomial is

$$|(M \setminus I) \cap K| + |(M \cap I) \setminus K|,$$

which vanishes only for $K = I$ among sets containing M^c . Hence the chosen coefficient uniquely dominates and makes (2) negative for a finite t . Viewed as the characteristic polynomial at a new spectral variable μ , the same determinant is negative at $\mu = \lambda$ and positive for sufficiently large μ . It therefore has a real zero $\mu > \lambda > 0$. $\square$

The substitution $\lambda = 0$ is not valid in general. For

$$J = \begin{pmatrix} -10 & 20 & -50 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}, \quad M = \{1\},$$

the matrix J is Hurwitz, the signed determinant of the only proper block containing the immobile set $\{2, 3\}$ is positive, but

$$J - \text{diag}(257/2, 0, 0)$$

has eigenvalue $3/2$. At that positive spectral value,

$$\det(\tfrac{3}{2}I_2 - J[\{2, 3\}, \{2, 3\}]) = -\tfrac{1}{4}.$$

This example separates Theorem 2.3 from any zero-parameter principal-minor rule.

3 Conservation laws and nonzero spatial modes

Let $S \subseteq \mathbb{R}^n$ be the stoichiometric subspace and suppose $J(\mathbb{R}^n) \subseteq S$, as occurs for a mass-action Jacobian $J = \Gamma M$ with $S = \text{im } \Gamma$.

Proposition 3.1 (Semisimple conservation block). *If $J(S) \subseteq S$ and the restriction $J|_S$ is Hurwitz, then*

$$\mathbb{R}^n = S \oplus \ker J, \quad J \sim \begin{pmatrix} J|_S & 0 \\ 0 & 0 \end{pmatrix}.$$

In particular, all conservation-induced zero eigenvalues are semisimple and, with $s = \dim S$,

$$\det(\lambda I - J) = \lambda^{n-s} \det(\lambda I_s - J|_S) > 0 \quad (\lambda > 0).$$

Proof. Hurwitz stability makes $J|_S$ invertible, hence $J(S) = S$. Because the full image of J is contained in S , this forces $\text{im } J = S$ and $\text{rank } J = s$. If $x \in S \cap \ker J$, invertibility of the restriction gives $x = 0$. Rank-nullity now gives $\dim S + \dim \ker J = n$, proving the direct sum. Both summands are invariant, and J vanishes on its kernel, which gives the block representation. The determinant factorization follows, and its second factor is positive on the positive real axis because it is the characteristic polynomial of a Hurwitz matrix. $\square$

Choose any rational basis matrix $B \in \mathbb{Q}^{n \times s}$ for S and any rational left inverse C with $CB = I_s$. Since JB has columns in S ,

$$JB = B(CJB).$$

Thus $R = CJB$ represents $J|_S$. If $\tilde{B} = BT$ is another basis, then a compatible left inverse gives

$$\tilde{R} = T^{-1}RT,$$

so all reduced Hurwitz tests are basis-independent.

3.1 Why a nonzero mode is not restricted to S

For Neumann boundary conditions on a connected bounded domain, every nonconstant Laplacian eigenfunction ϕ is orthogonal to the constant function, hence

$$\int_{\Omega} \phi(x) dx = 0.$$

A linear perturbation of the form $u(x) = a\phi(x)$ therefore satisfies every integrated linear conservation law $\ell^T \int_{\Omega} u = 0$ for every amplitude $a \in \mathbb{R}^n$. The amplitude space of a nonzero mode is the full species space, and its matrix is $J - q^2 D$.

Restricting that matrix to S is generally impossible. For example, with $S = \text{span}(1, 1)^T$ and $D = \text{diag}(1, 2)$,

$$D(1, 1)^T = (1, 2)^T \notin S.$$

The homogeneous reaction matrix preserves S ; unequal species diffusion need not. This distinction is essential for conservative networks.

4 Mass-action parametrization and the exact decision formula

Consider indexed reactions $y_r \rightarrow y'_r$, with source matrix $Y = [y_1 \cdots y_m]$ and stoichiometric matrix $\Gamma = [y'_1 - y_1 \cdots y'_m - y_m]$. At a positive state x^* , put

$$v_r = k_r (x^*)^{y_r}, \quad h_i = (x_i^*)^{-1}.$$

The equilibrium equation is $\Gamma v = 0$. Differentiating $k_r x^{y_r}$ at x^* gives the row vector $v_r y_r^T \text{diag}(h)$, and hence

$$J(v, h) = \Gamma \text{diag}(v) Y^T \text{diag}(h). \quad (3)$$

Conversely, if $v > 0$, $\Gamma v = 0$, and $h > 0$, choose

$$x_i^* = h_i^{-1}, \quad k_r = \frac{v_r}{(x^*)^{y_r}}.$$

These quantities are positive, x^* is an equilibrium, and direct differentiation recovers (3). Thus the parametrization is onto the set of positive-equilibrium Jacobians; it also handles the zero complex, parallel indexed reactions, autocatalysis, and arbitrary molecularity without modification.

Set

$$A(v) = \Gamma \text{diag}(v) Y^T, \quad J(v, h) = A(v) \text{diag}(h).$$

For $I \subseteq [n]$, column scaling gives

$$(-1)^{|I|} \det J(v, h)[I, I] = \left(\prod_{i \in I} h_i \right) (-1)^{|I|} \det A(v)[I, I]. \quad (4)$$

Let B and C be rational matrices with columns of B forming a basis of $S = \text{im } \Gamma$ and $CB = I$. Define

$$R(v, h) = C J(v, h) B.$$

By Section 3, this is a coordinate matrix for $J(v, h)|_S$ and changes only by similarity when the rational basis changes.

Theorem 4.1 (Exact all-mobile characterization). *A finite classical mass-action network admits a positive equilibrium that is Hurwitz on S and a strictly positive all-species diffusion matrix producing a positive real nonzero-mode eigenvalue if and only if there are $v \in \mathbb{R}_{>0}^m$, $h \in \mathbb{R}_{>0}^n$, and a nonempty $I \subseteq [n]$ such that*

$$\Gamma v = 0, \quad (5)$$

$$R(v, h) \text{ is Hurwitz}, \quad (6)$$

$$(-1)^{|I|} \det A(v)[I, I] < 0. \quad (7)$$

Proof. For necessity, begin with positive rates and equilibrium. The construction above gives positive v, h satisfying (5), while relative homogeneous stability is exactly (6). The spatial positive real eigenvalue and Theorem 2.1 give a negative signed principal minor of J . Equation (4) removes h from its sign and gives (7). Under relative Hurwitz stability the empty and full sets cannot be witnesses, so a nonempty branch is sufficient.

For sufficiency, reconstruct x^* and k from v, h . Condition (6) gives homogeneous stability on the compatibility class. Equations (4) and (7) give a negative signed principal minor of the full Jacobian. Theorem 2.1 supplies $D \succ 0$ and a positive real eigenvalue of $J - D$. Rescaling D by any chosen positive Neumann eigenvalue realizes it as $q^2 D_{\text{phys}}$ for a nonzero spatial mode. The amplitude is unrestricted by conservation, as proved above. $\square$

Routh–Hurwitz determinants of the characteristic polynomial of $R(v, h)$ turn (6) into finitely many strict rational polynomial inequalities. Together with the linear equations and positivity, Theorem 4.1 is a finite semialgebraic formula computed from the indexed pair (Γ, Y) . It eliminates the original rate constants, the equilibrium concentrations, the diffusion coefficients, the domain scale, and the wave number, but it does not assert a first, simple, or transverse stationary crossing.

For a designated mobile set M , replace (7) by the finite disjunction

$$\exists \lambda > 0, \quad I \supseteq M^c : \quad \det(\lambda I_{|I|} - J(v, h)[I, I]) < 0.$$

Proposition 3.1 verifies the background positivity required by Theorem 2.3.

5 Polynomial-size selector encoding

The explicit principal-set disjunction has exponentially many branches when n varies. Introduce variables b_i satisfying $b_i^2 - b_i = 0$ and put $S_b = \text{diag}(b_1, \dots, b_n)$. If $I = \{i : b_i = 1\}$, simultaneously permute rows and columns so that I comes first. The matrix

$$I - S_b - S_b A S_b$$

then has block form

$$\begin{pmatrix} -A[I, I] & 0 \\ 0 & I_{I^c} \end{pmatrix}.$$

Consequently

$$\det(I - S_b - S_b A S_b) = (-1)^{|I|} \det A[I, I]. \quad (8)$$

The empty selector gives determinant 1, so it cannot satisfy a strict negative inequality.

A determinant of an $n \times n$ matrix has a division-free arithmetic circuit of polynomial size, as do characteristic coefficients and the Hurwitz minors. Rational row reduction computes a basis B of $\text{im } \Gamma$ and a left inverse C with polynomial encoding length: all numerators and denominators are

ratios of minors of the rational input matrix. Replacing every arithmetic gate by a fresh variable and one polynomial equation gives a sparse existential formula of polynomial total size. Positivity and negativity are legitimate atomic relations in $\exists\mathbb{R}$; alternatively, they can be encoded by

$$g > 0 \iff \exists z (gz^2 - 1 = 0), \quad g < 0 \iff \exists z ((-g)z^2 - 1 = 0).$$

Thus the all-mobile decision problem belongs to the existential theory of the reals. This is only a membership result; no $\exists\mathbb{R}$ -hardness claim is used.

6 The open-cube Partition family

Let $a_1, \dots, a_\ell$ be positive integers. Choose $k = \lceil \sqrt{\ell + 1} \rceil$ and $m = k^2 > \ell$, and pad the input by zeros to $a \in \mathbb{Z}^m$. Put

$$\gamma = a^T a, \quad Q = I_m + aa^T, \quad \beta = 1 - \frac{1}{2m(1 + \gamma)},$$

and define

$$B(x, y) = \begin{pmatrix} -kQ & y \\ x^T & k\beta \end{pmatrix}, \quad x, y \in (-1, 1)^m. \quad (\text{OC})$$

Theorem 6.1 (Exact open-cube equivalence). *The family (OC) contains a Hurwitz matrix if and only if the original PARTITION instance has signs $t_i \in \{-1, 1\}$ satisfying $\sum_{i=1}^\ell a_i t_i = 0$. A YES instance has an explicit rational witness strictly inside the cube.*

Proof. Suppose $t \in \{-1, 1\}^m$ satisfies $a^T t = 0$; signs on padded zero entries are arbitrary. Set

$$r = \frac{1 + \beta}{2} \in (0, 1), \quad x = rt, \quad y = -rt.$$

Since $r^2 - \beta = (1 - \beta)^2/4 > 0$, the coordinates are strictly interior. The vectors a and t are orthogonal and $\|t\| = \sqrt{m} = k$. On $\text{span}(a)$ the top-left block has eigenvalue $-k(1 + \gamma)$; on the orthogonal complement of $\text{span}(a, t)$ it has eigenvalue $-k$. With $u = t/k$, the remaining invariant two-dimensional block, in the basis $(u, 0), (0, 1)$, is

$$\begin{pmatrix} -k & -rk \\ rk & k\beta \end{pmatrix}.$$

Its trace is $k(\beta - 1) < 0$ and determinant $k^2(r^2 - \beta) > 0$. It is Hurwitz, as are all complementary blocks.

Conversely, suppose $B(x, y)$ is Hurwitz. At $(x, y) = (0, 0)$ the matrix has m negative eigenvalues and one positive eigenvalue, whereas a Hurwitz $(m + 1) \times (m + 1)$ matrix has positive signed determinant $(-1)^{m+1} \det B$. Therefore the signed determinant changes sign along $B(\theta x, \theta y)$, $0 \leq \theta \leq 1$, and a singular point occurs at some $\theta_0 \in (0, 1)$. Write $z = -y$. The Schur complement of $-kQ$ gives

$$\theta_0^2 x^T Q^{-1} z = k^2 \beta = m\beta,$$

so $x^T Q^{-1} z > m\beta$.

For $M = Q^{-1} \succ 0$,

$$2x^T M z \leq x^T M x + z^T M z.$$

A convex quadratic reaches its maximum over a cube at a vertex, and equality in the bilinear upper bound is attained by taking the same maximizing vertex for x and z . Hence

$$\max_{x,z \in [-1,1]^m} x^T Q^{-1} z = \max_{t \in \{-1,1\}^m} t^T Q^{-1} t.$$

Sherman–Morrison gives

$$Q^{-1} = I_m - \frac{aa^T}{1 + \gamma},$$

and therefore

$$\max_t t^T Q^{-1} t = m - \frac{\min_t (a^T t)^2}{1 + \gamma}.$$

Because $m\beta = m - 1/[2(1 + \gamma)]$, the strict inequality above implies $\min_t (a^T t)^2 < 1/2$. The scalar $a^T t$ is an integer, so it is zero. The first ℓ signs solve the original instance. $\square$

The construction has polynomial encoding length: k and m are polynomial in ℓ , and the entries of Q and β have bit length polynomial in the binary input length. The padding introduces no new partition constraint.

7 A one-parameter-per-row stability lift

Write a rational rank-one affine family as

$$B(q) = B_0 + \sum_{r=1}^d q_r u_r w_r^T = B_0 + UX(q),$$

where $U = [u_1 \ \cdots \ u_d]$ and the r th row of $X(q)$ is $q_r w_r^T$. For rational $\alpha > 0$, define

$$L(q) = \begin{pmatrix} B_0 & U \\ X(q)(B_0 + \alpha I) & X(q)U - \alpha I_d \end{pmatrix}. \quad (\text{SL})$$

Lemma 7.1 (Exact row-splitting similarity). *With*

$$P(q) = \begin{pmatrix} I & 0 \\ -X(q) & I \end{pmatrix}, \quad P(q)^{-1} = \begin{pmatrix} I & 0 \\ X(q) & I \end{pmatrix},$$

one has

$$P(q)L(q)P(q)^{-1} = \begin{pmatrix} B(q) & U \\ 0 & -\alpha I_d \end{pmatrix}. \quad (\text{SIM})$$

Thus $L(q)$ is Hurwitz exactly when $B(q)$ is Hurwitz, and $\det L(q) = (-\alpha)^d \det B(q)$. Every top row is fixed; lower row r is of the form $a_r + q_r b_r$ and contains no parameter q_s with $s \neq r$.

Proof. Right multiplication first gives

$$L(q)P(q)^{-1} = \begin{pmatrix} B_0 + UX(q) & U \\ X(q)(B_0 + UX(q)) & X(q)U - \alpha I_d \end{pmatrix}.$$

Subtracting $X(q)$ times the top block row from the lower block row gives (SIM). Block triangularity proves the spectral and determinant assertions. The fixed part of lower row r is $-\alpha e_r^T$ in the last d coordinates, while its variable part is

$$q_r (w_r^T (B_0 + \alpha I) \mid w_r^T U),$$

which proves the rowwise parameter claim and checks all dimensions and signs. $\square$

8 Eliminating arbitrary positive right scalings in the reduction family

Apply the lift of Section 7 with $\alpha = 1$ to the $2m$ varying entries of the open-cube family. Thus $q = (y, x)$ and the lift has order $3m + 1$.

Theorem 8.1 (Scaling elimination). *For the lifted PARTITION family,*

$$\exists q \in (-1, 1)^{2m}, \exists D \succ 0 \text{ diagonal} : L(q)D \text{ is Hurwitz}$$

holds if and only if the PARTITION instance is YES.

Proof. A partition witness gives a Hurwitz $B(q)$ by Theorem 6.1; the similarity (SIM) makes $L(q)$ Hurwitz, so $D = I$ proves one direction.

For the converse suppose $L(q)D$ is Hurwitz, and split $D = \text{diag}(D_0, D_1)$ according to the $(m + 1) + 2m$ block sizes. At $q = 0$,

$$L(0)D = \begin{pmatrix} B_0 D_0 & U D_1 \\ 0 & -D_1 \end{pmatrix}, \quad B_0 = \text{diag}(-kQ, k\beta).$$

The first m coordinates of $B_0 D_0$ are similar to $-kD_{0,t}^{1/2} Q D_{0,t}^{1/2} \prec 0$, while its last coordinate is the positive scalar $k\beta d_* > 0$. The lower block $-D_1$ is negative definite. Consequently $L(0)D$ has exactly one positive and all remaining negative real eigenvalues. Its signed determinant $(-1)^{3m+1} \det(L(0)D)$ is negative; for the Hurwitz endpoint $L(q)D$ the same signed determinant is positive. Continuity supplies $\theta_0 \in (0, 1)$ such that $\det(L(\theta_0 q)D) = 0$.

The positive diagonal D is nonsingular, so $\det L(\theta_0 q) = 0$. The row-splitting determinant identity transfers this singularity to $B(\theta_0 q)$. Writing $z = -y$, its Schur complement yields

$$\theta_0^2 x^T Q^{-1} z = m\beta.$$

The strict endpoint relation $\theta_0 < 1$ implies $x^T Q^{-1} z > m\beta$. The cube-maximization and integrality argument in Theorem 6.1 then produces a partition sign vector. Notice that the relative magnitudes of D were used only to establish the one-positive-eigenvalue inertia at the initial endpoint and disappear from the singularity equation; they cannot evade the argument. $\square$

9 Exact mass-action realization of independent row segments

Let $p \geq 1$. Split the row indices into fixed rows F and variable rows V . For $i \in F$ prescribe $a_i \in \mathbb{Q}^{1 \times p}$; for $i \in V$ prescribe $a_i, b_i \in \mathbb{Q}^{1 \times p}$. Define $L(q)$ by

$$L_i(q) = a_i \quad (i \in F), \quad L_i(q) = a_i + q_i b_i \quad (i \in V), \quad -1 < q_i < 1.$$

Theorem 9.1 (Row-segment realization). *There is an indexed classical mass-action network on exactly p species, with nonnegative integer source and target complexes and stoichiometric columns in $\{\pm e_1, \dots, \pm e_p\}$, whose complete positive steady-flux Jacobian-factor image is*

$$\{\Gamma \text{diag}(v) Y^T : v > 0, \Gamma v = 0\} = \{RL(q) : R \succ 0 \text{ diagonal}, q \in (-1, 1)^{|V|}\}. \quad (\text{RS})$$

The network has full stoichiometric rank. It uses $2|F| + 3|V|$ reactions, and its encoding length is polynomial in the rational input length.

Proof. Fix a variable row i . Choose an integer $c_i > 0$ clearing the denominators of both $a_i + b_i$ and $a_i - b_i$. Choose a nonnegative integer base complex y_i^0 with $(y_i^0)_i \geq 1$ and large enough componentwise that

$$y_i^+ = y_i^0 + c_i(a_i + b_i), \quad y_i^- = y_i^0 + c_i(a_i - b_i)$$

are nonnegative integer vectors. Add the three indexed reactions



Their stoichiometric columns are $e_i, e_i, -e_i$. If their positive fluxes are u_i, w_i, s_i , the i th steady-flux equation is exactly $s_i = u_i + w_i$. Their contribution to row i of $\Gamma \text{diag}(v)Y^T$ is therefore

$$\begin{aligned} u_i(y_i^+)^T + w_i(y_i^-)^T - (u_i + w_i)(y_i^0)^T &= c_i((u_i + w_i)a_i + (u_i - w_i)b_i) \\ &= \rho_i(a_i + q_i b_i), \end{aligned}$$

where

$$\rho_i = c_i(u_i + w_i) > 0, \quad q_i = \frac{u_i - w_i}{u_i + w_i} \in (-1, 1).$$

Conversely, every $\rho_i > 0$ and $q_i \in (-1, 1)$ is attained by

$$u_i = \frac{\rho_i(1 + q_i)}{2c_i}, \quad w_i = \frac{\rho_i(1 - q_i)}{2c_i}, \quad s_i = \frac{\rho_i}{c_i},$$

which are all strictly positive.

For a fixed row choose $c_i > 0$ clearing the denominators of a_i , choose $y_i^0 \geq 0$ with $(y_i^0)_i \geq 1$ and $y_i^+ = y_i^0 + c_i a_i \geq 0$, and add



The steady-flux equation forces the two positive fluxes to be equal. Giving each the value ρ_i/c_i produces row $\rho_i a_i$, and every positive steady flux in this row has that form.

All columns belonging to row i are scalar multiples of e_i . Hence the global equation $\Gamma v = 0$ is precisely the collection of the independent row equations just described; there are no hidden cross-row constraints. This proves both inclusions in (RS), including zero row entries and repeated or parallel indexed reactions. Each row contains both e_i and $-e_i$, so Γ has row rank p . Sources and targets are distinct because every reaction changes coordinate i by one. Denominator clearing and taking a componentwise maximum plus one construct the bases with polynomial bit length, proving the complexity assertion. $\square$

At $x^* = \mathbf{1}$, the rate constants are exactly the chosen positive fluxes. Thus the construction uses no zero rates and no limiting argument.

10 NP-hardness for classical mass action

Theorem 10.1 (Weak NP-hardness). *Weak all-mobile stationary-Turing admissibility of rationally encoded finite classical mass-action networks is NP-hard under polynomial-time many-one reductions from PARTITION. For an input of length ℓ , the construction chooses $m = k^2 > \ell$ and has $3m + 1$ species and $8m + 2$ indexed reactions. Every stoichiometric column is $\pm e_i$, the stoichiometric rank is full, and all rates and equilibrium coordinates in a YES witness are strictly positive.*

Proof. Form the lifted matrix family $L(q)$ from the open-cube instance. It has $m + 1$ fixed rows and $2m$ variable rows. Apply Theorem 9.1; the complete positive steady-flux image is

$$A(v) = RL(q), \quad R \succ 0 \text{ diagonal}, \quad q \in (-1, 1)^{2m}.$$

At an arbitrary positive equilibrium, every Jacobian is

$$J = RL(q)H, \quad H \succ 0 \text{ diagonal}.$$

Conjugation by R gives

$$R^{-1}JR = L(q)HR = L(q)D, \quad D = HR \succ 0.$$

The scaling-elimination theorem therefore says that some positive-equilibrium Jacobian is Hurwitz exactly when the partition instance is YES; no theorem about unrestricted diagonal stabilizability is invoked.

The coordinate inherited from the bottom-right entry of the open-cube matrix is a fixed row and column of the lift, with $L_{jj} = k\beta > 0$. Hence every realized Jacobian has

$$J_{jj} = \rho_j k \beta h_j > 0,$$

so its signed singleton minor is $-J_{jj} < 0$. Whenever the homogeneous Jacobian is Hurwitz, the fixed-J diagonal-damping theorem supplies a strictly positive diffusion matrix producing a positive real nonzero-mode eigenvalue. Conversely, admissibility includes homogeneous stability. Thus the reaction network is stationary-Turing-admissible exactly when the partition instance is YES.

For a YES instance choose the rational interior witness of the open-cube proof, $R = H = I$, $x^* = \mathbf{1}$, and rates equal to the strictly positive row-realization fluxes. There are

$$2(m + 1) + 3(2m) = 8m + 2$$

reactions. Denominator clearing and the base-complex construction have polynomial bit complexity. This is a polynomial-time many-one reduction. $\square$

The theorem is ordinary (weak) NP-hardness because the source problem is PARTITION. The source and target complexes produced by denominator clearing can have molecularity growing with the input; no bounded-molecularity hardness statement follows from this construction.

11 Polynomial-time exact decision for fixed species count

Fix n and define

$$K = \{v \in \mathbb{R}_{\geq 0}^m : \Gamma v = 0\}, \quad \Phi(v) = \Gamma \operatorname{diag}(v) Y^T, \quad P = \Phi(K).$$

Lemma 11.1 (Relative-interior projection). *If K contains a strictly positive vector, then*

$$\{\Phi(v) : v > 0, \Gamma v = 0\} = \operatorname{ri} P.$$

Proof. A coordinate inequality $v_r \geq 0$ is not identically tight on K because a strictly positive point exists. Hence $\operatorname{ri} K = K \cap \mathbb{R}_{> 0}^m$. For any nonempty convex set and linear map T , one has $T(\operatorname{ri} C) = \operatorname{ri} T(C)$; this follows by restricting T to the affine hull and using the finite-dimensional open-mapping theorem. Apply this with $T = \Phi$. $\square$

Lemma 11.2 (Circuit enumeration). *Every extreme ray of K has support at most $\text{rank } \Gamma + 1 \leq n + 1$. For fixed n , all extreme rays and their exact rational generators are computable in polynomial time in the reaction count and input bit length.*

Proof. An extreme ray is a support-minimal nonzero nonnegative dependency among the columns of Γ . If its support contained more than $\text{rank } \Gamma + 1$ columns, the restricted kernel would have dimension at least two. A nonproportional kernel perturbation can be scaled in both directions until one coordinate vanishes while retaining nonnegativity, decomposing the original vector into two nonproportional vectors of K , a contradiction. Thus every support is a positive circuit of size at most $n + 1$.

Enumerate all column subsets of size at most $n + 1$, compute their exact nullspaces, and retain the one-dimensional dependencies with a same-sign representative whose support is minimal. This includes singleton zero columns and tolerates repeated columns. There are $O(m^{n+1})$ subsets. Circuit coordinates can be chosen as signed minors of matrices of order at most n ; for fixed n , determinant bounds give polynomial encoding length. $\square$

Theorem 11.3 (Fixed-species complexity). *For every fixed number of species n , weak all-mobile stationary-Turing admissibility of rationally encoded classical mass-action reaction networks is decidable in polynomial bit complexity. The same is true for each fixed designated mobile set.*

Proof. First decide whether K has a strictly positive point by rational linear programming; by homogeneity this is equivalent to feasibility of $\Gamma v = 0$, $v_r \geq 1$. If not, no positive equilibrium exists.

Map every circuit generator from Lemma 11.2 through Φ . Their images generate P completely. The ambient dimension is n^2 , fixed. In fixed dimension, exact generator-to-relative-facet conversion is polynomial: compute the span and lineality, quotient by the lineality, enumerate candidate supporting hyperplanes from bounded-size subsets of generators, and test them against all generators. Determinantal normal vectors have polynomial bit length, and the fixed-dimensional upper bound makes the number of candidates polynomial. Lifting back gives

$$E \text{vec}(A) = 0, \quad \ell_j(A) > 0$$

for $A \in \text{ri } P$. This description also covers lower-dimensional and nonpointed images, $P = \{0\}$, and the case where the quotient has no facets.

By Lemma 11.1, admissibility is equivalent to an existential formula in the n^2 entries of A and the n positive entries of h : the relative-facet conditions above, the Routh–Hurwitz conditions for $A \text{diag}(h)$ restricted to $S = \text{im } \Gamma$, and one negative signed principal minor of A . A rational basis of S and left inverse have polynomial bit length by minor bounds. For fixed n , the number of real variables and all determinant and Hurwitz degrees are fixed, while the number and coefficient lengths of input polynomials are polynomial. Fixed-variable algorithms for the existential theory of the reals therefore run in polynomial bit complexity.

For a designated mobile set, add one real variable λ and the relevant principal-block determinant condition. The variable count remains fixed. $\square$

This is an algorithmic theorem, not a finite motif or topology classification. For $n = 1, 2, 3$ it gives complete polynomial-time exact decision algorithms for arbitrary reaction lists.

12 Finite exact certificate completeness

Fix one principal-set branch of the semialgebraic characterization. Replace every strict inequality by an equation:

$$g > 0 \iff \exists z (gz^2 - 1 = 0), \quad g < 0 \iff \exists z ((-g)z^2 - 1 = 0). \quad (\text{EQ})$$

These equivalences are exact over the reals: a solution forces the indicated sign, and a strict sign supplies $z = 1/\sqrt{|g|}$.

Theorem 12.1 (Two-sided finite certificates). *Every YES instance of the exact all-mobile decision problem has a finite real-algebraic sample certificate. Every NO instance has a finite family of Real Nullstellensatz identities, one for each nonempty principal branch (or, equivalently, one identity for the Boolean-selector encoding). Both kinds are independently checkable by exact arithmetic. No polynomial bound on their size is asserted.*

Proof. A nonempty semialgebraic set defined over $\mathbb{Q}$ contains a point with real-algebraic coordinates. The finitely many coordinates generate a finite real extension of $\mathbb{Q}$, which is simple: represent it as $\mathbb{Q}(\alpha)$ using the irreducible minimal polynomial of α , a rational isolating interval selecting the real embedding, and polynomial representatives for every coordinate. Exact reduction modulo the minimal polynomial checks all equations; Sturm or subresultant sign determination at the isolated root checks every required sign. This gives a finite YES certificate tied to the reconstructed network formula rather than to an unresolved symbolic parameter system.

For an infeasible branch, let $F_1, \dots, F_N$ be the polynomial equations after (EQ). The Real Nullstellensatz states that emptiness of their real zero set is equivalent to the existence of polynomials q_k and s_j with

$$-1 = \sum_j s_j^2 + \sum_{k=1}^N q_k F_k. \quad (\text{RN})$$

A verifier expands both sides and compares every rational coefficient. At a hypothetical common real zero, (RN) would give $-1 = \sum_j s_j^2 \geq 0$, proving soundness. There are finitely many principal branches, so a NO instance has a finite collection of such identities. $\square$

Existence, verification, and generation must be distinguished. Identities and algebraic samples can in principle be generated by exhaustive enumeration of degrees, rational coefficients, minimal polynomials, isolating intervals, and exact identity checks; termination follows from the theorem. The result does not provide a polynomial certificate-size bound or a practical general certificate generator. A bundled classifier may therefore return UNRESOLVED on a raw network without contradicting the decision or certificate-existence theorems.

13 Exact data-level separating examples

13.1 Unsigned and signed interaction graphs

Consider

$$J_0 = \begin{pmatrix} -7 & -5 & -4 \\ -7 & -7 & -5 \\ -3 & -4 & -6 \end{pmatrix}, \quad J_1 = \begin{pmatrix} -7 & -1 & -2 \\ -7 & -4 & -4 \\ -3 & -7 & -6 \end{pmatrix}.$$

Both have the same complete all-negative signed interaction graph. Their characteristic polynomials are respectively

$$\lambda^3 + 20\lambda^2 + 66\lambda + 47, \quad \lambda^3 + 17\lambda^2 + 53\lambda + 16,$$

and both satisfy the cubic Hurwitz inequalities. Every signed principal minor of J_0 is positive. The $(2, 3)$ principal determinant of J_1 is -4 , and

$$D = \text{diag}(9, 1/9, 1/9), \quad \det(D - J_1) = -281/81.$$

Apply the fixed-row case of the row realization theorem. Each network has the same ordered stoichiometric matrix

$$[e_1, -e_1, e_2, -e_2, e_3, -e_3]$$

and all positive-equilibrium Jacobians are RJ_kH with $R, H \succ 0$ diagonal. For every principal set the signed determinant is multiplied by a positive factor. Thus the first network can never be stationary-destabilized, while the second is admissible at $R = H = I$. Levels 1 and 2 therefore fail for the network-wide existential problem.

13.2 The stoichiometric matrix alone

Both two-species networks use

$$\Gamma = [e_1, -e_1, e_2, -e_2].$$

For the first, take sources

$$(2, 0), \quad (1, 2), \quad (2, 0), \quad (0, 3)$$

and add the corresponding stoichiometric column to form each target. Unit steady flux gives

$$A = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix},$$

which is Hurwitz and has a positive first diagonal entry. For the second, use sources

$$(1, 0), \quad (1, 0), \quad (0, 1), \quad (0, 1).$$

The steady-flux equations pair the positive and negative reaction in each row, and their source contributions cancel, so $A(v) = 0$ for every positive steady flux. The two networks have the same indexed Γ and opposite answers.

Level 3 is a Coates graph of one selected Jacobian and is not a parameter-independent network representation. Level 5, the indexed pair (Γ, Y) , is sufficient by the exact formula. Level 6 determines and is determined by Level 5 when reaction indices and integer labels are retained.

14 Independent software architecture

File	Independent role
<code>network_parser.py</code>	Parses indexed source and target complexes and reconstructs Y and Γ .
<code>mass_action_jacobian.py</code>	Differentiates mass-action monomials directly and separately evaluates the flux factorization.
<code>partition_reduction.py</code>	Builds the open-cube family, row-splitting lift, exact similarity, and Hurwitz determinants.
<code>fixed_n_projection.py</code>	Enumerates positive circuits using nullspaces and an independent cofactor implementation; maps and decomposes the projected cone.
<code>certificate_checks.py</code>	Checks rational and real-algebraic sample data and polynomial identities with mutations.

15 Adversarial test record

The final exact campaign includes the following independently generated tests.

Campaign	Count	Coverage
Fixed-J complete bounded census	20,317 matrices	All 1×1 entries in $[-4, 4]$, all 2×2 entries in $[-2, 2]$, and all 3×3 matrices over $\{-1, 0, 1\}$.
Fixed-J random rational	5,000 matrices	Negative-minor search, exact multiscale determinant witness, numerical root corroboration.
Conservation	600	Rank-deficient invariant subspaces, direct-sum and determinant factorization checks.
Boolean selectors	30,480	Every selector for 840 random small matrices.
Mass-action row families	490	2,940 exact positive steady-flux samples, both image inclusions.
Partition instances	917	Exhaustive sign enumeration for lengths 2–6 and entries 1–6; 334 YES and 583 NO.
Row-splitting exact checks	32	Symbolic similarity and characteristic-polynomial comparison.
Fixed- n circuit comparisons	1,159	Nullspace and cofactor implementations agree.
Exact cone decompositions	3,644	Zero/repeated columns, lower-dimensional and zero images, conservation, and parallel reactions.
Mobile theorem	1,500	Exact determinant expansions plus 120 constructed necessity checks and the zero-parameter counterexample.

Numerical searches were used only to seek counterexamples. The retained claims do not depend on failure of a numerical optimizer to find one.

16 Certificate formats and software limitation

A YES certificate contains the reconstructed network, one principal branch, and all coordinates in one explicitly isolated real algebraic number field. The verifier checks the canonical equations and signs from the network data. A NO certificate contains a Real Nullstellensatz identity for every branch (or one selector-encoded branch). Identity checking is exact polynomial arithmetic.

The release does not vendor a complete cylindrical algebraic decomposition or Nullstellensatz generator. Consequently the command-line classifier may return **UNRESOLVED** for a raw network without a supplied certificate. This is an implementation limitation, not a claim that failed search proves a NO instance.

17 Strengthening routes not claimed

1. The row-realization sources have molecularity growing with the binary input. Parallel bounded reactions would be pseudo-polynomial, and no exact finite-rate auxiliary-species gadget was proved.
2. The source problem is PARTITION; no strongly NP-hard interval family was obtained.
3. No exact addition/multiplication gadget library established $\exists\mathbb{R}$ -hardness.
4. A bounded numerical campaign found generic simple stationary first crossings along the constructed paths, but this neither proves universal primary-crossing equivalence nor excludes another path with an earlier wave crossing.
5. The fixed- n code implements circuit projection and exact preprocessing, not a complete practical general CAD solver.

18 Reproducibility commands

From the Phase III project root:

```
bash release/one_command_replay.sh
python -m unittest discover -s independent_verifier/tests -v
python red_team/partition_reduction_falsifier.py
python red_team/fixed_n_falsifier.py
python external_audit/minimal_verifier.py external_audit/yes_partition_1_1.json
```

The replay also verifies that both inherited archives retain their original SHA-256 values, compiles both manuscripts and the external-audit PDFs, runs all exact tests, renders the PDFs, and checks the final manifest.